

IZHAAN MUBEEN

📍 Lahore, Pakistan

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Objective

Mechatronics and Control Engineering undergraduate with experience in **robotic mechanism design**, **SolidWorks CAD modeling**, and **MATLAB/Simscape Multibody simulation**. Skilled in developing **robotic assemblies**, **multibody models**, and CAD-to-prototype workflows for advanced robotic systems.

Education

University of Engineering and Technology, Lahore	2024 – Present
B.Sc. Mechatronics & Control Engineering	CGPA: 3.34/4.0
Government College University, Lahore	2022 – 2024
Intermediate (Pre-Engineering) – Grade: A	

Research & Engineering Experience

SolidWorks Design Intern 2026
Lahore, Pakistan

- Designed and developed **mechanical assemblies and mechanisms** for robotic and industrial applications using **SolidWorks**, focusing on structural design, component integration, and manufacturability.
- Created detailed **3D CAD models**, assemblies, and engineering drawings from design concepts and physical measurements.
- Performed **motion studies** and structural simulations to evaluate mechanical behavior and improve design reliability.
- Developed fabrication-oriented designs including weldments and mechanical components for **prototype development**.

Master Bootcamp Organizing Team – BRIDGE Project (Co-Funded by European Union) 2026
UET, Lahore

- Served on the management team for the Master Bootcamp, part of the **BRIDGE Project**, a collaboration co-funded by the **European Union** hosted at **UET, Lahore**, handling photography, videography, and event coverage.
- Assisted with general event management and coordination alongside faculty and participants throughout the bootcamp.

Projects

VISTA: Mobile Service Robot Mechanical Design

SolidWorks, Mechanical Design, Robotic Mechanisms

- Developed the **mechanical architecture** of a mobile service robot by preparing initial design drawings and structural layouts for the robot base.
- Designed the robot base and upper torso assembly including tapered housing, rotating head mechanism, and **shoulder-mounted revolute joints and linkage structures**.
- Modeled mechanical components considering **joint integration**, workspace requirements, range of motion, and fabrication constraints.
- Prepared detailed **CAD models** and design documentation to support future **physical prototyping** and validation.

Dynamic Modeling and Control of a Differential Drive Mobile Robot

MATLAB, Simulink, Simscape Multibody, Robotics Simulation

- Developed a physics-based digital model of a **differential-drive robot** by importing a URDF-based robot description into **MATLAB Simscape Multibody**.
- Integrated **CAD-derived STL geometries**, rigid-body properties, inertia parameters, coordinate transformations, and joint constraints for **multibody simulation**.
- Implemented wheel actuation through revolute joints and developed **differential-drive control logic** for translational and rotational motion.
- Analyzed simulated robot behavior by evaluating **joint motion responses** under different control inputs.

Automated Conveyor Belt Sorting System

SolidWorks, Motion Study, Mechanical Simulation

- Designed an automated sorting mechanism using **parametric CAD modeling** with mechanical linkages and actuated components.
- Performed motion studies and structural analysis to evaluate mechanism operation and reliability.

Technical Skills

CAD & Mechanical Design: SolidWorks (3D Modeling, Assembly Design, Engineering Drawings, Motion Study, SolidWorks Simulation), Mechanical Prototyping

Simulation & Modeling: MATLAB, Simulink, Simscape Multibody, URDF-Based Robot Modeling, Multibody Dynamics

Robotics: Robotic Mechanisms, Kinematics, Differential Drive Systems, ROS2

Programming: MATLAB, Python, C, Assembly

Tools: Solidworks, Keil uVision, Proteus, VS Code